

Vehicle Features — Functional Guide

BLE App + IoT Tracker · Observable Behaviour Specification

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Document control

Field	Value
Document	Vehicle Features — Functional Guide
Audience	Customers and integrators using the BLE app and IoT tracker
Scope	Describes <i>observable behaviour</i> , not implementation
Platform	EC200U IoT tracker · companion MCU · BLE mobile app
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Note on scope. This document specifies what the system *does* as seen from outside — what a user, integrator or support engineer can observe. Implementation detail lives in the internal project context document.

1. System at a glance

The IoT tracker is the centre of the system. It is powered by the **auxiliary battery and is always awake**. The **main battery** is a separate unit that identifies itself over CAN.

Bluetooth and the motion sensor never switch off. They are how the vehicle is woken and how it is protected, so every power-saving behaviour in this document is designed around that constraint rather than overriding it.

Signal map

From	To	Carries
BLE App	IoT Tracker	Lock / Unlock / Boot / Ignition commands
IoT Tracker	Companion MCU	Relayed commands (boot release)
Companion MCU	IoT Tracker	Wheel-sensor voltage

Main Battery	IoT Tracker	CAN 0x540 heartbeat
Motion Sensor	IoT Tracker	Movement and tamper events
IoT Tracker	Cloud	Telemetry, when online
IoT Tracker	Nearby Apple devices	Find My beacon → Find My network
IoT Tracker	Outputs	Buzzer · Wheel lock · Ignition relay

2. Power modes

Three modes, chosen automatically to save power on a parked vehicle.

Mode	What works	What is off	Typical use
Active	Everything — tracking, cloud, CAN, app	—	Riding / in use
Derated	App, motion sensing, battery link	GPS, cloud / mobile data	Vehicle stirred but not in use
Low Power	App, motion sensing	GPS, cloud / mobile data, battery link	Parked and undisturbed

Power draw drops roughly 4× in Low Power compared with a fully connected device.

Transitions

From	To	Trigger
Active	Low Power	Ignition OFF for 2 minutes
Low Power	Derated	Small movement detected
Derated	Low Power	No movement for 2 minutes
Low Power	Active	App command or significant motion
Derated	Active	App command or theft detected

What the user notices

- The app **always connects**, in every mode.
- In Derated and Low Power the vehicle **stops appearing online**; live location is stale.

- Sending any app command returns the vehicle to Active. Reconnecting to the network takes **approximately 20–40 seconds**, so the first cloud update after waking is not instant.
- Recovery is immediate — the device never sleeps, so there is no boot delay.

3. BLE app commands

Command	Effect	Refused when
Lock	Short beep; anti-theft armed	Ignition is ON
Unlock / Ignition-off	If ignition ON, turns it off ; otherwise double beep and disarms anti-theft	Never
Boot unlock	Opens the boot (relayed to the companion MCU)	Companion MCU unreachable
Ignition ON	Energises the ignition relay	Main battery not detected, or vehicle is locked

Every command returns an **acknowledgement** to the app — ACK ... on success or NAK ... with the reason on refusal — so a refusal is always visible rather than silent.

Buzzer and the ignition circuit. The buzzer is powered from the ignition circuit, so the device briefly raises ignition to make lock and unlock beeps audible. This is **not** an ignition-on event: the vehicle stays logically off.

4. Main-battery detection and ignition control

The main battery announces itself with a repeating CAN heartbeat. Detection is **asymmetric on purpose** — quick to trust, slow to distrust.

Condition	Result
Heartbeat received	PRESENT — immediately, one heartbeat is enough
Silent for 10 seconds	ABSENT for power logic
Silent for 30 seconds	ABSENT for ignition

Consequences

- **Present is instant.** One heartbeat is enough.
- **Absent needs sustained silence**, so a single dropped CAN message never causes a false “battery removed”.
- Ignition **cannot be switched on** without a live main battery.
- If the battery is removed **while riding**, ignition is **switched off automatically** after 30 seconds. No app command is needed.

- If ignition is already off when the battery goes absent, ignition stays off and any new Ignition-ON request is refused.

5. Anti-theft

Armed by **Lock**, disarmed by **Unlock**. Fully automatic once armed — no app interaction required.

Detectors

Three independent detectors. Any one of them triggers the alarm.

Detector	Detects
Wheel-sensor movement	Vehicle being wheeled or rolled
Significant motion	Vehicle lifted or carried
Small nudge	Tamper attempt on a stationary vehicle

Each detector has a debounce, so a one-off bump or vibration does **not** raise an alarm.

Alarm behaviour

- The alarm runs in **10-second cycles** and repeats for as long as tampering continues.
- While active it drives the **buzzer**, the **wheel lock** and the **ignition line**.
- **Unlock always wins** — it stops the alarm immediately.
- Detection keeps working even if the motion sensor fails, because the wheel sensor is a **separate channel**.
- Anti-theft is active in **every power mode**, including Low Power.

6. Behaviour after a power cut or reset

On reset, outputs start in a safe state with everything off. What the device then restores differs per item, because the risk differs per item.

Item	Survives reset?	Why
Locked / unlocked state	Yes	A reset must not silently unlock a parked vehicle
Ignition on/off	No — always off	Safety: a relay must only energise on a fresh, deliberate command
Power mode	No — starts Active	Safest default; re-enters Low Power normally

So a vehicle locked before a power interruption is **still locked and still guarded** afterwards, and anti-theft re-arms automatically. A rider whose vehicle resets must **re-send Ignition ON**.

7. Find My — locating the vehicle without mobile network

The tracker continuously broadcasts an **Apple “Find My” (OpenHaystack) beacon** over Bluetooth. Any passing iPhone or iPad silently relays the sighting to Apple’s Find My network, where **only the owner** can decrypt the location.

Recovery path

Step	Actor	Action
1	Vehicle (no SIM / no coverage)	Broadcasts BLE beacon
2	Passer-by’s iPhone	Relays sighting silently
3	Apple Find My network	Stores encrypted report
4	Owner	Decrypts — the only party who can

Why it matters

- Works with **no SIM, no mobile coverage and no cloud connection**. It needs only a nearby Apple device.
- Runs in **every power mode**, including Low Power, because Bluetooth never switches off.
- It is a **last-resort locator**: if a thief removes the main battery, or the vehicle is in a basement with no signal, the beacon keeps advertising on the auxiliary battery.

How it coexists with the app

The device advertises the Find My beacon *and* keeps its normal VEC-<IMEI> name and connectable services at the same time — so the BLE app still finds, connects to and controls the vehicle exactly as before. The Bluetooth address is derived from the device’s Find My key, so each vehicle appears as its own distinct tag.

Provisioning requirement. Each unit needs its **own unique Find My key**.

Devices sharing one key are indistinguishable in the Find My network and cannot be located individually. This is a manufacturing-line requirement, not a firmware one.

8. How it all works together

Bluetooth, motion sensing and anti-theft (when locked) are always available. Power mode controls only GPS, mobile data, cloud and the main-battery link.

Interaction	Rule
Anti-theft vs. power saving	Anti-theft outranks power saving. An alarm can occur in any mode, and temporarily drives the ignition line to sound the buzzer.
Ignition vs. power mode	Ignition depends on the main-battery link, which is switched off in Low Power. The app command wakes the device first so the link is restored — but the vehicle must be awake for ignition to be permitted.
Lock vs. power mode	Independent. Locking does not change the power mode, and the power mode does not disarm anti-theft.
Motion as shared trigger	A small nudge lifts Low Power to Derated; significant motion returns the device to Active <i>and</i> can raise an alarm if the vehicle is locked.

Design principles behind these rules

1. **The device never truly sleeps.** It sheds radios instead. This costs a little standby current versus a true sleep state and buys zero boot delay on recovery, plus anti-theft that is live in Low Power.
2. **Asymmetric thresholds where the two errors have different costs.** Battery present versus absent, false alarm versus missed tamper, lock state versus ignition state — each is timed to the cost of getting it wrong in that direction.
3. **No silent failures.** Every refused command names its reason; every degraded mode is visible to the user rather than hidden behind a spinner.